

Lecture 13

Adversarial

(x, y_{true}) — object

$f_{\theta}(x)$ — neural network

$\mathcal{L}(\theta, x, y)$ — loss value

$$f_{\theta}(x) = y_{true}$$

$$\tilde{x} \approx x : f_{\theta}(\tilde{x}) \neq y_{true}$$

$$\|\tilde{x} - x\|_{\infty} < \frac{\epsilon}{255}$$

ϵ — attack budget

$$\max_i |\tilde{x}_i - x_i|$$

- White box — gradient is available
- Black box — gradient not available

• Targeted attack $f_{\theta}(\tilde{x}) = y_{target}$

• Untargeted attack $f_{\theta}(\tilde{x}) \neq y_{true}$

FGSM (Fast Gradient Sign Method)

— untargeted

$$\tilde{x} = x + \epsilon \cdot \text{sign}[\nabla_x \mathcal{L}(\theta, x, y_{true})]$$

— targeted

$$\tilde{x} = x - \epsilon \cdot \text{sign}[\nabla_x \mathcal{L}(\theta, x, y_{target})]$$

iFGSM (Iterative FGSM)

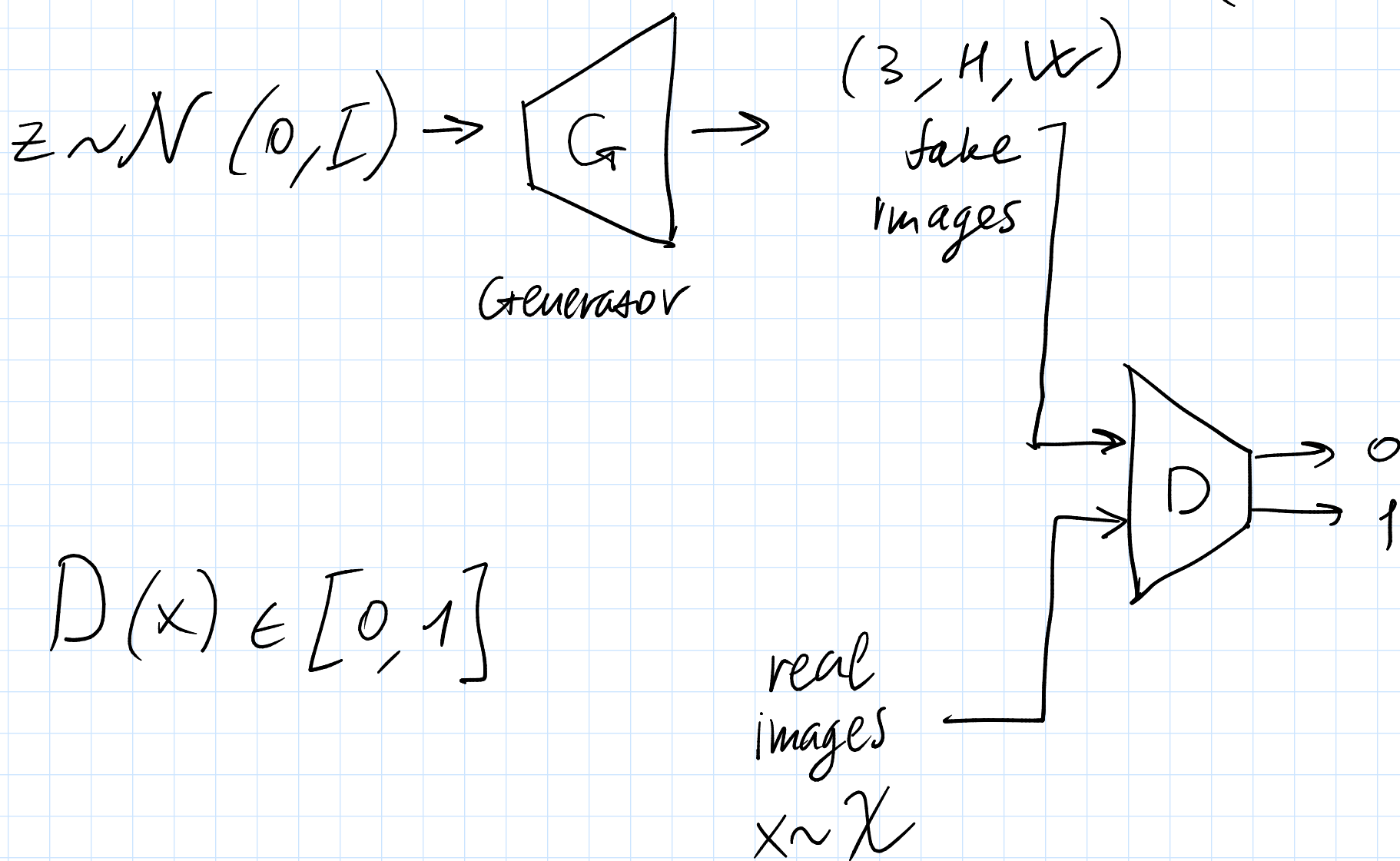
$$\tilde{x}_0 = x$$

for $i = 1 \dots N$:

$$\tilde{x}_i = \tilde{x}_{i-1} + \frac{\epsilon}{N} \text{sign}(\nabla_x h(\theta, \tilde{x}_{i-1}, y_{\text{true}}))$$

$$\tilde{x} = \tilde{x}_N$$

Generative Adversarial Networks (GANs)



$$D(x) \in [0, 1]$$

$$\min_G \max_D \left\{ \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim \mathcal{N}(0, I)} [\log (1 - D(G(z)))] \right\}$$

Style GAN

LS GAN, WGAN, Cramer

cGAN (Conditional GAN)

$$G(z, c)$$

$$D_y^1(x) - \text{head for GAN loss}$$

$$D_y^2(x) - \text{head for classification}$$

Image to Image

$P_{1 \times 2} P_{1 \times}$

((Cycle GAN))

$$x_s \sim p_s(x) \quad x_T \sim p_T(x)$$

$$G(x_s) \approx x_T$$

$$\text{MSE loss} : \mathbb{E}_{x_s, x_T} \|G(x_s) - x_T\|_2^2 \rightarrow \min_G$$

$$\text{GAN loss} : \mathcal{L}_{\text{GAN}}(G, D)$$

$$\text{MSE + GAN} : \alpha \mathcal{L}_{\text{MSE}} + (1 - \alpha) \mathcal{L}_{\text{GAN}}$$

$$\mathcal{L}_{\text{GAN}} = \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{x \sim p_{\text{gen}}} [\log(1 - D(x))] \rightarrow \max_D$$

$x = G(z)$

$$D^*(x) - ?$$

$$\mathcal{L}_{\text{GAN}} = \int_{x \in \mathcal{X}} [p_{\text{data}}(x) \log D(x) + p_{\text{gen}}(x) \log(1 - D(x))] dx$$

$$g(c) = p_{\text{data}}(x) \log c + p_{\text{gen}}(x) \log(1-c)$$

$$D^*(x) \rightarrow c^*$$

$$g'(c) = \frac{p_{\text{data}}(x)}{c} - \frac{p_{\text{gen}}(x)}{1-c} = 0$$

$$(1-c) p_{\text{data}} - c p_{\text{gen}}(x) = 0$$

$$D^*(x) = c = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_{\text{gen}}(x)} \in [0, 1]$$

$$L_{\text{GAN}}^* = \int \left[p_{\text{data}}(x) \log \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_{\text{gen}}(x)} + \right.$$

$$\left. + p_{\text{gen}}(x) \log \frac{p_{\text{gen}}(x)}{p_{\text{data}}(x) + p_{\text{gen}}(x)} + p_{\text{data}}(x) \log 2 - \right.$$

$$\left. - p_{\text{data}}(x) \log 2 + p_{\text{gen}}(x) \log 2 - p_{\text{gen}}(x) \log 2 \right] dx =$$

$$= \int \left[p_{\text{data}}(x) \log \frac{2 p_{\text{data}}(x)}{p_{\text{data}}(x) + p_{\text{gen}}(x)} + \right.$$

$$\left. + p_{\text{gen}}(x) \log \frac{2 p_{\text{gen}}(x)}{p_{\text{data}}(x) + p_{\text{gen}}(x)} \right] dx -$$

$$- \log 2 \int (p_{\text{data}}(x) + p_{\text{gen}}(x)) dx =$$

$$= KL \left(p_{\text{data}} \parallel \frac{p_{\text{data}} + p_{\text{gen}}}{2} \right) + KL \left(p_{\text{gen}} \parallel \frac{p_{\text{data}} + p_{\text{gen}}}{2} \right)$$

$$- 2 \log 2 = 2 JS(p_{\text{data}}, p_{\text{gen}}) - 2 \log 2$$

Mode collapsing

